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X-CBNet: An Explainable Effective Deep Learning Framework Based on Spectrograms for Predicting Valvular Disorder using PCG Signals

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Abstract

Valvular Heart Disease (VHD), caused by malfunctioning heart valves, poses significant diagnostic challenges due to the complexity of heart sound patterns and variability in clinical presentations. Traditional auscultation methods are subjective, and existing automated models often function as black boxes, offering limited insight into the reasoning behind predictions. Therefore, there is a pressing need for accurate, interpretable, and multi-class diagnostic tools to aid clinicians in early and reliable detection of VHD using phonocardiogram (PCG) signals. X-CBNet, a convolutional neural network (CNN) followed by a bidirectional long short-term memory (Bi-LSTM) network framework, is employed in this research to harness deep learning's capability to achieve high diagnostic accuracy while ensuring interpretability through explainable AI methods. Melspectrograms are used to capture essential features of the phonocardiograms. The proposed model is designed as a five-class classifier distinguishing between aortic stenosis, mitral stenosis, mitral regurgitation, mitral valve prolapse, and normal heart sounds. Gradient-weighted class activation mapping (Grad-CAM) is utilized for explainability, generating heatmaps that visualize model decision-making. The proposed X-CBNet model achieved an overall accuracy of 99.15%. Per-class accuracies are: Normal 99.15%, Aortic Stenosis 98.43%, Mitral Regurgitation 98.79%, Mitral Stenosis 99.58%, and Mitral Valve Prolapse 99.80%. The model also achieved a macro-average AUC of 0.99 and a micro-average AUC of 0.98, while maintaining a misclassification rate of less than 1%. The Grad-CAM heatmaps further validate model decisions by highlighting class-discriminative features. The robust performance demonstrates that the proposed model is suitable for real-time clinical applications, offering enhanced transparency through the integration of explainable AI.

Keywords Explainable-AI · Phonocardiogram signals · Spectrogram · Grad-CAM · Valvular heart disease

Introduction

The narrowing, blockage, or impaired function of arteries within the cardiovascular system characterizes cardiovascular disease (CVD). These abnormal arteries hinder the heart's ability to circulate oxygenated blood throughout the body effectively [17, 19]. Due to its unpredictable course and heightened mortality rates, cardiovascular disease poses a significant global threat [23]. The incidence of mortality

due to cardiac arrest has shown a significant rise, especially among older demographics. United Nations projections suggest that the elderly population is set to double by the year 2050 [28].

In past times, medical professionals utilized specialized tools such as the 'stethoscope' to diagnose conditions like cardiac arrest through direct listening to patients' heartbeat sounds. However, these diagnostic methods lacked precision and systematic validation, posing ethical concerns regarding their scientific validity [13]. The subjective nature inherent in traditional procedures presents a challenge, as interpretations can vary among physicians. Accurate identification of cardiac abnormalities is crucial, given that 90% of heart diseases can be preventable by managing risk factors. Therefore, early detection of cardiac issues plays a pivotal role in averting severe consequences. Developing computer-assisted tools for analyzing heart sound signals is urgently needed. These signals contain early pathological indicators

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linked to cardiovascular disorders and have demonstrated efficacy in early disease prediction. Clinical experts can manually extract disease-specific features from PCG waveforms to aid in diagnosing various cardiac conditions.

Heart sounds generated by the opening and closing of valves are brief and fleeting. Typically, heart sound recordings are obtained from four main areas: the aortic, pulmonary, tricuspid, and mitral. Identifying abnormal factors in the PCG signal indicates the presence of VHDs. S1 and S2 are regular, and S3 and S4 are irregular sounds, like murmurs, captured by phonocardiography, as shown in Fig. 1. Detecting the S3 and S4 sounds in PCG signals is crucial for diagnosing systolic and diastolic heart failures, representing significant pathological symptoms [25].

Heart valve abnormalities typically manifest in two primary forms: "stenosis," characterized by narrowing a heart valve that restricts blood flow, and "regurgitation," where the valve fails to prevent blood from flowing backwards. VHDs encompass a range of cardiovascular conditions stemming from abnormalities in one or more heart valves. These conditions include aortic regurgitation (AR), aortic stenosis (AS), mitral regurgitation (MR), mitral stenosis (MS), and mitral valve prolapse (MVP). These disorders result from improper opening and closing of heart valves throughout the cardiac cycle [10].

The cardiac cycle comprises all the heart events, from initiating one heartbeat to starting the next. This cycle includes two phases characterized by different modes of contraction and relaxation called systole and diastole. The first heart sound (S1) originates from the functioning of the tricuspid and mitral valves. The opening and closing of the aortic

and pulmonary valves produce the second heart sound (S2). Abnormal heart sounds, such as the third (S3) and fourth (S4) heart sounds, may result from conditions like congenital disabilities between the left and right ventricles, valve prolapse, obstruction, or calcification. Automated analysis of heart sounds in clinical applications typically involves steps depicted in Fig. 2.

The typical procedure for computer-based analysis of heart sounds involves several key steps [12]:

- Preprocessing (segmentation and filter)
- Feature extraction
- Classifier design

Before analysis, heart sound recordings undergo preprocessing steps to assess signal integrity, including baseline correction and high-frequency noise filtration. Segmentation defines the boundaries for each heartbeat phase (S1, systolic, S2, diastolic), facilitating the extraction of pertinent features. Subsequently, classification techniques correlate these features with identified phases, or heart sounds within each segmented beat. Recent advancements have seen the emergence of multiple methodologies for automatically detecting valvular heart diseases using PCG signals.

Deep learning (DL) models have recently shown significant potential in automated VHD diagnosis through PCG signals. Several studies have highlighted the importance of PCG signal processing for cardiac disease diagnosis, including applications in noise removal, segmentation, and clinical decision support [1]. However, a critical drawback

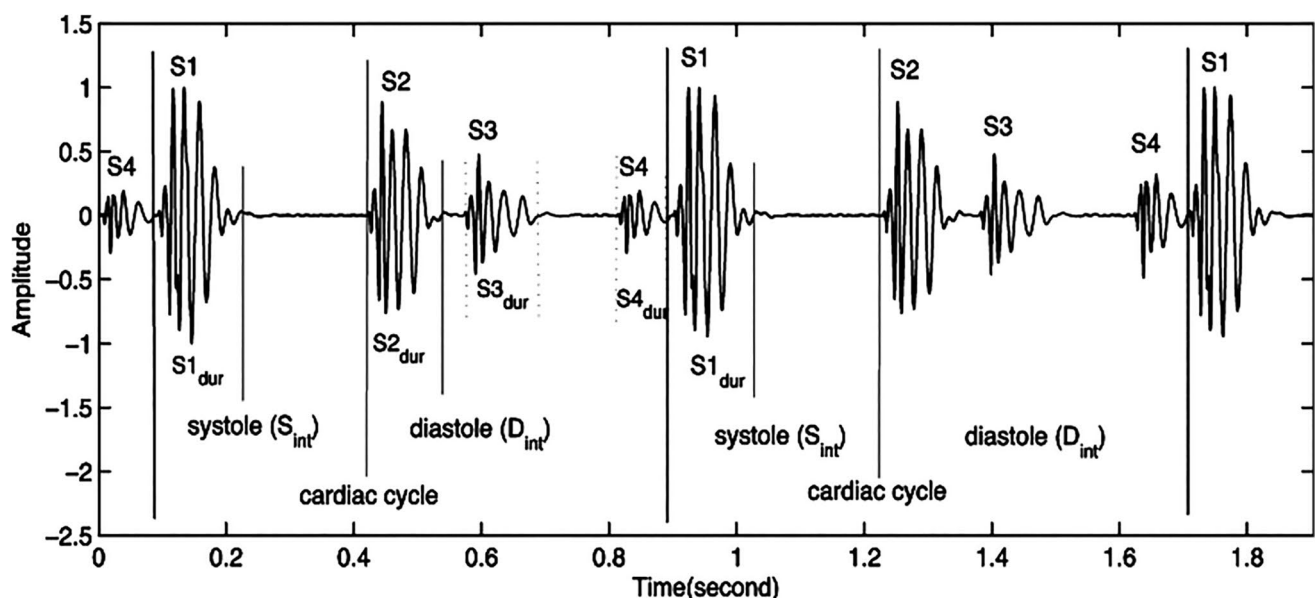
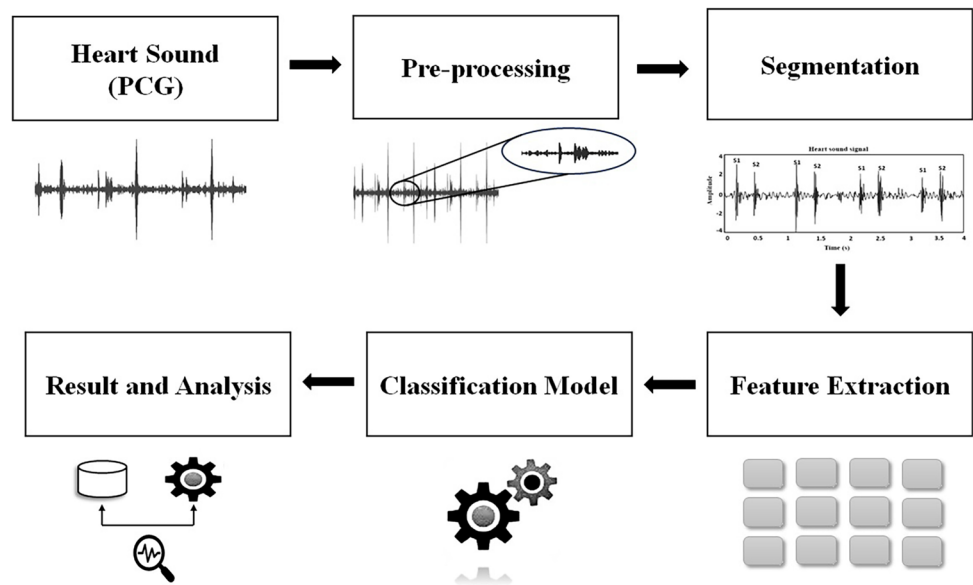


Fig. 1 An example of the cardiac cycle in PCG signal

Fig. 2 General workflow of detection of valvular heart disease using PCG signals



of these models is their need for interpretability, which hinders their seamless integration into clinical practice. There is a pressing need for systems that achieve high accuracy and provide clear interpretations of their predictions. The absence of explainable artificial intelligence (XAI) [22] reduces adoption and trust among medical professionals despite the various models proposed for VHD diagnosis. Due to their interpretability limitations, many of these models need to be more suitable for real-time diagnosis. The key contributions and principal advancements introduced by this research include:

- Enhancing the transparency of DL model diagnostic conclusions by applying eXplainable AI (XAI).
- The interpretability of the CNN-BiLSTM model's transparency improved in detecting cardiac abnormalities, employing a novel application of Grad-CAM for PCG signals.
- A comprehensive assessment is performed to analyze the multi-class classification of VHD. The performance of the proposed X-CBNet model is compared against existing deep learning classifiers.

The article is structured as follows: Section [Related Work](#) reviews existing literature on VHDs and XAI implementations. Section [Proposed Methodology](#) details the methodology employed in this study. Section [Results and Discussion](#) presents the computational outcomes related to VHD diagnosis and Grad-CAM interpretation. Lastly, Section [Conclusion](#) summarizes the key findings of the research and discusses their implications for future studies in this field.

Related Work

The efficacy of machine learning methods in identifying VHDs relies on extracting features from PCG signals and applying feature selection techniques. DL approaches have recently gained prominence for VHD detection using PCG signals, bypassing the need for extensive feature evaluation. Temporal DL models such as CNNs and recurrent neural networks (RNNs) have been employed to differentiate between normal and pathological heart sounds [9, 29].

A comprehensive deep learning framework (LU-Net) is employed to simultaneously reduce ambient and internal lung sound interferences, enabling real-time denoising of heart sounds [3]. Sundar et al. [27] classified five classes for heart sounds utilizing various machine learning classifiers. The LSTM classifier attained the highest accuracy with 94.5%.

A novel approach integrates a CNN based on deep learning with Cardi-Net using power spectrograms. This method aims to extract highly discriminative features from PCG signals, facilitating the identification of diseases [15]. The study [5] evaluated the effectiveness of transfer learning in detecting heart murmurs, assessing three models: VGG16, VGG19, and ResNet50.

The authors [6], integrated machine learning techniques along with discrete fourier transform (DFT) features from PCG signals to autonomously detect VHDs. Researchers have employed discrete wavelet transforms (DWT) and Mel frequency cepstral coefficient (MFCC) [20] based features in conjunction with diverse machine learning techniques to classify valvular heart diseases from PCG signals [30]. Three distinct cardiac sound characteristics Mel spectral

mapping, MFCCs, and sub band envelope are extracted and evaluated using diverse feature fusion and segmentation approaches with feedforward CNNs [7].

Despite numerous studies on heart sound classification, many have encountered reduced performance due to the complex and variable nature of cardiac acoustic environments. Early detection of cardiac disorders is crucial for enabling patients to take preventive measures against potential harm. This study introduces a novel and effective explainable deep-learning framework utilizing spectrograms and Grad-Cam to predict valvular disorders.

Proposed Methodology

This section presents a comprehensive methodology to implement a reliable and interpretable deep learning framework. This framework leverages spectrograms derived from PCG signals to forecast valvular disorders, thereby supporting clinical decision-making with enhanced insights and the help of a prediction explainability approach.

Data Curation

A publicly accessible 2016 PhysioNet/CinC challenge PCG signal dataset is employed to devise an interpretable deep-learning framework to predict valvular heart disease [18]. The dataset includes phonocardiogram recordings collected from 764 subjects in diverse clinical and non-clinical environments, compiled by seven distinct research groups. The dataset includes 3240 PCG recordings featuring 84,425 cardiac cycles with heart rates ranging from 35 to 159 bpm. Recordings vary from 5 to 120 seconds and originate from six clinical settings. Besides variations in acquisition devices and recording setups, the recordings include various noises, such as breathing sounds, stethoscope movements, intestinal activity, and peripheral talking, which pose challenges for developing noise-robust algorithms. The samples of the PCG dataset are shown in Fig. 3.

Data Processing

The denoising model has been trained using cardiac sound recordings sourced from the dataset featured in the 2016 PhysioNet/CinC challenge. All recordings underwent resampling to 1kHz, and a third-order Butterworth bandpass filter with a passband ranging from 25Hz to 400Hz was applied. Segments with exceptionally high amplitudes were identified and removed to mitigate noisy spikes, thereby reducing background noise interference. Normalization is performed to standardize signal amplitudes within the range

of $[-1, 1]$. Some initial segments of recordings contained significant noise; therefore, the cleanest 5-second segments are extracted based on spectral amplitude and wavelet entropy considerations.

The crucial initial stage of the classification algorithm involves preprocessing the cardiac sound signals. A Butterworth filter is designed to have a maximally flat frequency response in the passband. A bandpass filter allows frequencies within a certain range to pass while attenuating frequencies outside. This filter eliminates unwanted components such as baseline drift, high-frequency interference, and artefacts from the captured initially cardiac sound signals. After filtering, the heart sound signals are analyzed based on amplitude and temporal characteristics. The transfer function $H(s)$ of a Butterworth filter of order n is given by Eq. 1:

$$H(s) = \frac{1}{\sqrt{1 + \left(\frac{s}{\omega_c}\right)^{2n}}} \quad (1)$$

where, s is the complex frequency variable, and ω_c is the cutoff frequency.

For a bandpass filter, we need two cutoff frequencies ω_1 and ω_2 , representing the lower and upper cutoff frequencies, respectively. The transformation from a low-pass prototype to a bandpass filter is given by Eq. 2:

$$s \rightarrow \frac{s^2 + \omega_0^2}{Bs} \quad (2)$$

where, $\omega_0 = \sqrt{\omega_1\omega_2}$ is the center frequency, and $B = \omega_2 - \omega_1$ is the bandwidth. For a fifth-order Butterworth filter, the poles of the transfer function in the s -plane are shown in Eq. 3:

$$s_k = \omega_c e^{j\frac{(2k-1)\pi}{2n}}, \quad k = 1, 2, \dots, n \quad (3)$$

For a bandpass filter, the poles are mapped using the bandpass transformation as shown in Eq. 4:

$$s_k = \frac{s^2 + \omega_0^2}{Bs} \quad (4)$$

The transfer function of the fifth-order Butterworth bandpass filter is shown in Eq. 5:

$$H(s) = \frac{1}{\sqrt{1 + \left(\frac{s^2 + \omega_0^2}{Bs\omega_c}\right)^{10}}} \quad (5)$$

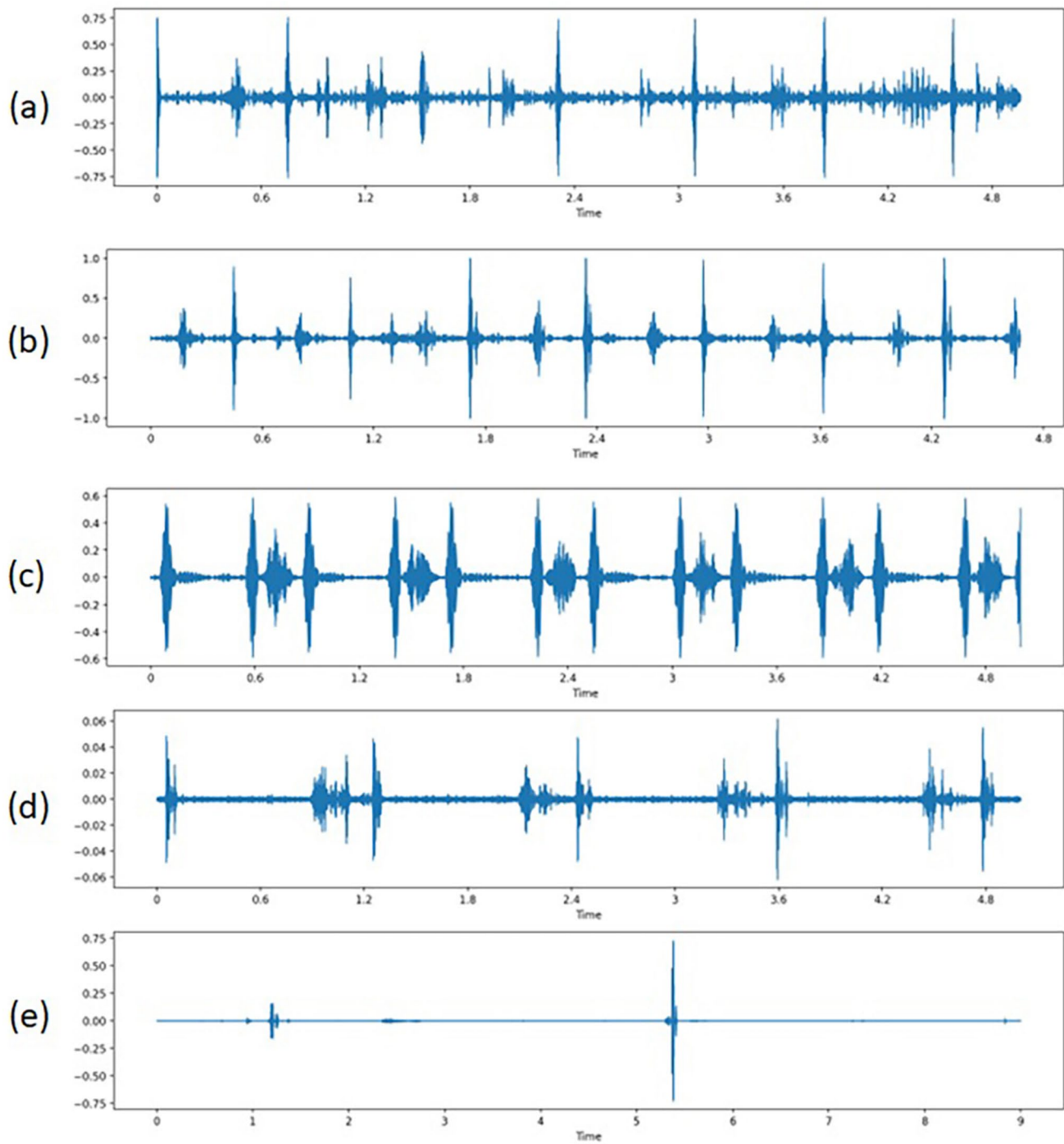


Fig. 3 PCG sound signal classes (a) normal (b) AS (c) MR (d) MS (e) MVP

Simplifying this, ω_0 and B can be substituted into the Eq. 6:

$$H(s) = \frac{1}{\sqrt{1 + \left(\frac{s^2 + (2\pi \times \sqrt{25 \times 400})^2}{2\pi \times 375 \times s} \right)^2}} \quad (6)$$

In practice, applying the Butterworth filter involves transforming the analog filter into its digital equivalent using techniques like bilinear transformation or impulse invariance. The filter coefficients are then applied to the discrete signal.

Given a discrete heart sound signal $x[n]$, the output $y[n]$ after applying the bandpass filter can be represented as shown in Eq. 7:

$$y[n] = \sum_{k=0}^N b_k x[n-k] - \sum_{k=1}^N a_k y[n-k] \quad (7)$$

where, b_k and a_k are the filter coefficients determined by the filter design process.

The filtering process includes the removal of baseline drift in which the low-frequency components below 25 Hz are removed. The frequencies above 400 Hz are attenuated and unwanted noise and artifacts are filtered out. After filtering, the heart sound signals are analyzed for amplitude and temporal characteristics, which include the peak amplitude, and root mean square (RMS) amplitude and duration of heart sounds, time intervals between sounds (e.g., S1 and S2). These characteristics are crucial for further classification algorithms in detecting abnormalities in cardiac sound signals.

Window Length Optimization

Choosing the appropriate window lengths for processing is a critical system parameter in a real-time framework designed for denoising heart sounds. Optimal window size balances latency and denoising efficacy, where larger sizes may introduce latency while shorter ones may compromise denoising performance. The Signal-to-Noise Ratio (SNR) improvement (ΔSNR) can be used to measure denoising efficacy. SNR is defined as the ratio of signal power (P_{signal}) to noise power (P_{noise}) as denoted in Eq. 8:

$$\text{SNR} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right) \quad (8)$$

When processing a window of length w , the SNR improvement can be defined shown in Eq. 9:

$$\Delta\text{SNR}(w) = \text{SNR}_{\text{out}}(w) - \text{SNR}_{\text{in}} \quad (9)$$

where, SNR_{in} is the initial SNR of the noisy heart sound, and $\text{SNR}_{\text{out}}(w)$ is the SNR after denoising using window length w . Latency (L) is the time delay introduced by processing. For a window-based processing system, the latency is directly proportional to the window length w as shown in Eq. 10:

$$L(w) = k \cdot w \quad (10)$$

where, k is a proportionality constant that includes factors such as computational overhead and buffer processing time. To find the optimal window length, we need to balance SNR improvement and latency. This can be formulated as an optimization problem where we maximize SNR improvement while minimizing latency. A cost function $J(w)$ that combines both aspects can be defined as shown in Eq. 11:

$$J(w) = \alpha \cdot \Delta\text{SNR}(w) - \beta \cdot L(w) \quad (11)$$

where, α and β are weighting factors that determine the relative importance of SNR improvement and latency, respectively.

The optimal window length of 0.8s is derived by experimentally evaluating the SNR improvements for different window lengths and balancing it against the introduced latency. The window length that maximizes the cost function $J(w)$ while considering both SNR improvement and latency is selected as the optimal parameter for the real-time denoising system.

Feature Extraction from PCG Signals

The cepstrum represents a signal by separating its source and filter parameters into two distinct additive components. It is calculated by performing the inverse DFT on the logarithm of the magnitude spectrum of a frame [11]. This is represented in the Eq. 12.

$$C(n) = \mathcal{F}^{-1} \{ \log (|\mathcal{F}\{x(n)\}|) \} \quad (12)$$

where, $C(n)$: Cepstrum of the signal, $\mathcal{F}$: Fourier transform, $\mathcal{F}^{-1}$: Inverse Fourier transform, $x(n)$: Input signal, $\log$: Logarithm function, $|\cdot|$: Magnitude.

Transforming a signal from the time domain to the frequency domain involves replacing convolution with multiplication. Additionally, logarithms are used to convert multiplication steps into addition. Afterwards, the inverse DFT is applied, distinguishing between rapidly and slowly varying signal parts. MFCC are widely recognized features that characterize phonocardiogram signals. These coefficients leverage the understanding that low-frequency components of sound signals convey phonetically more

significant information to human perception than high-frequency components. The process of computing mel-cepstrum is shown in Fig. 4.

Model Architecture for the Prediction of VHD

In the proposed X-CBNet model, the CNN is used for feature extraction from the input data, and the BiLSTM is used for capturing temporal dependencies. The workflow of the proposed methodology to predict VHD using explainable CNN-BiLSTM is shown in Fig. 5.

The convolution operation can be derived as: Given an input sequence $\mathbf{X} = [x_1, x_2, \dots, x_T]$ where $x_t \in \mathbb{R}^d$, the convolution operation is defined as:

$$h_t = f(\mathbf{W} * \mathbf{x}_{t:t+k-1} + \mathbf{b}) \quad (13)$$

where:

- $\mathbf{W}$ is the filter (weight matrix) of size $k \times d$.
- $*$ denotes the convolution operation.
- $\mathbf{x}_{t:t+k-1}$ is a window of k input vectors.
- $\mathbf{b}$ is the bias term.
- f is a non-linear activation function (e.g., ReLU).

After convolution, a pooling operation (e.g., max pooling) is applied to reduce the dimensionality:

$$p_i = \max_{t \in \text{pooling region}} h_t \quad (14)$$

where p_i is the pooled feature.

An LSTM cell is defined by the following equations:

$$i_t = \sigma(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i) \quad (15)$$

$$f_t = \sigma(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f) \quad (16)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o) \quad (17)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_c \mathbf{x}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c) \quad (18)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (19)$$

where, $\mathbf{i}_t$, $\mathbf{f}_t$, $\mathbf{o}_t$ are the input, forget, and output gates. and $\mathbf{c}_t$, $\mathbf{h}_t$, σ , $\odot$ denotes the cell state, hidden state, sigmoid function and element-wise multiplication respectively.

In a BiLSTM, the input sequence is processed in both forward and backward directions. Let $\vec{\mathbf{h}}_t$ and $\overleftarrow{\mathbf{h}}_t$ denote the forward and backward hidden states:

$$\vec{\mathbf{h}}_t = \text{LSTM}_{\text{forward}}(\mathbf{x}_t, \vec{\mathbf{h}}_{t-1}) \quad (20)$$

$$\overleftarrow{\mathbf{h}}_t = \text{LSTM}_{\text{backward}}(\mathbf{x}_t, \overleftarrow{\mathbf{h}}_{t+1}) \quad (21)$$

The final output at time step t is the concatenation of the forward and backward hidden states:

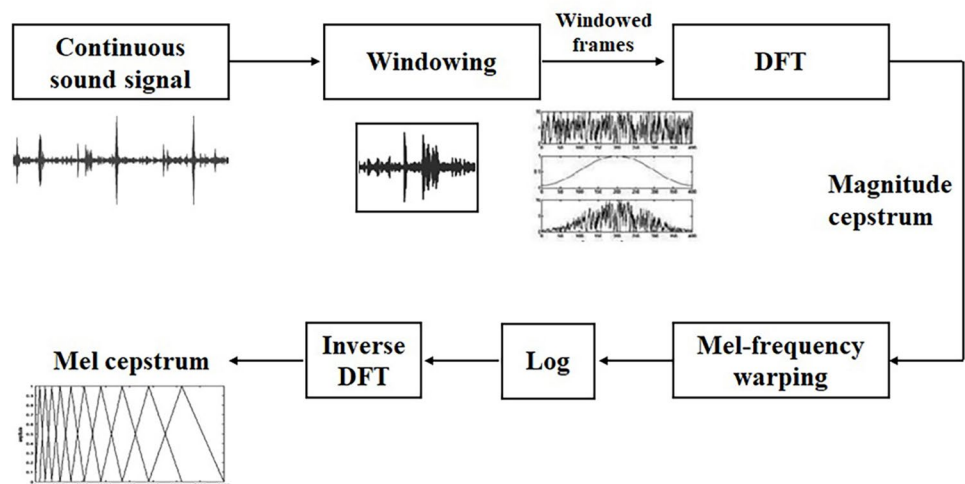
$$\mathbf{h}_t = [\vec{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t] \quad (22)$$

Combining the CNN and BiLSTM, the overall model is defined as follows:

1. Apply convolution and pooling to the input sequence $\mathbf{X}$ to obtain feature maps $\mathbf{P} = [p_1, p_2, \dots, p_m]$.
2. Pass the feature maps $\mathbf{P}$ through the BiLSTM to obtain the final hidden states $\mathbf{H} = [h_1, h_2, \dots, h_m]$.
3. Optionally, apply a fully connected layer and a softmax layer for classification.

The proposed model leverages the strengths of both CNNs for feature extraction and BiLSTMs for capturing temporal dependencies in sequential data. The proposed explainable

Fig. 4 Computation of mel-cepstrum



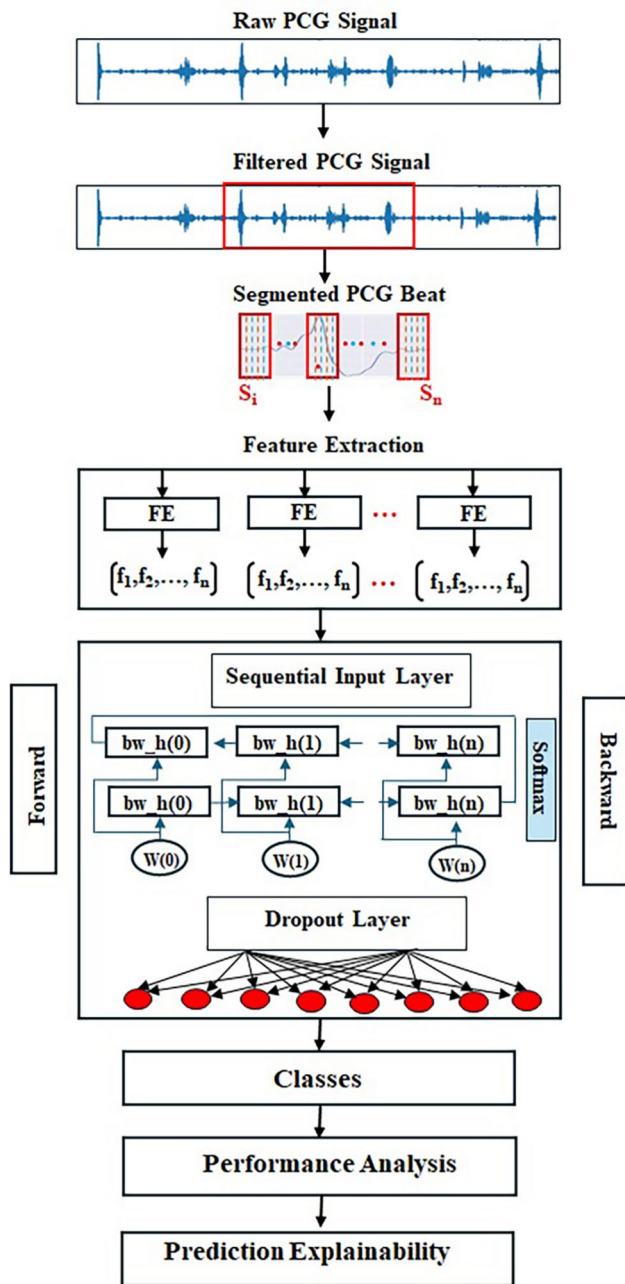


Fig. 5 Workflow to predict VHD

CNN-BiLSTM Model Layers for Predicting Valvular Heart Disease is shown in Table 1.

The input layer receives the input sequence of shape (sequence_length, num_features). The convolutional layer applies a number of filters to the input sequence to extract features. The pooling layer reduces the dimensionality of the feature maps, retaining the most important information. The first dropout layer is used to prevent overfitting by randomly setting a fraction of input units to 0. The bidirectional

LSTM layer captures temporal dependencies in both forward and backward directions. The flatten layer reshapes the output from the BiLSTM layer into a 1D vector. The dense layer applies a non-linear transformation to the input. The second dropout layer is used to prevent overfitting. The output layer produces the final classification probabilities. Table 2 summarizes the training configuration employed for developing the proposed X-CBNet model. The Adam optimizer was selected due to its adaptive learning capabilities, with a learning rate set to 0.001. A dropout rate of 0.3 is used to reduce overfitting, and categorical cross-entropy was chosen as the loss function for multi-class classification. The model was trained for 100 epochs with a batch size of 32, and early stopping was applied to prevent overtraining.

Prediction Explainability

In many cases, DL models are characterized by their opaque nature, leading to challenges in understanding the rationale behind their decisions [16]. Grad-CAM stands out among various XAI techniques designed to address this issue. It generates a heatmap based on the gradients flowing into the last convolutional layer of a deep neural network (DNN), helping to reveal which specific image features contribute most to the model's classification decisions. Grad-CAM relies on a global average pooling (GAP) layer to facilitate gradient retrieval from the final convolutional layer. Additionally, the ReLU activation in Grad-CAM ensures that the resulting heatmaps only highlight non-negative regions, enhancing the interpretability of the model's focus on critical image attributes.

Results and Discussion

This section provides comprehensive details on the results obtained from the experimental procedures, model training, and validation. It also includes a comparative performance analysis between the proposed technique and traditional approaches.

Simulation Environment

Python Jupyter Notebook 3.7.17 and Google Colab were utilized, along with libraries such as NumPy, SciPy, scikit-learn, and pandas. The tests were conducted on a system equipped with an i7-8565 Intel Core processor, 12 GB of RAM, and a 2 GB NVIDIA GeForce MX130 GPU, using the R2022a MATLAB programming environment. The experiments focused on multi-class classification tasks, with datasets evenly distributed among the tests.

Table 1 The proposed explainable CNN-BiLSTM model Layers for predicting VHD

Layer Type	Layer Name	Output Shape	Activation Function	Parameters
Input	Input Layer	(None, sequence_length, num_features)	-	-
Convolutional	Conv1D	(None, sequence_length - filter_size + 1, num_filters)	ReLU	filters=num_filters, kernel_size=filter_size
Pooling	MaxPooling1D	(None, pooled_length, num_filters)	-	pool_size=pool_size
Dropout	Dropout1	(None, pooled_length, num_filters)	-	rate=dropout_rate
Bidirectional LSTM	BiLSTM	(None, pooled_length, 2 * lstm_units)	-	units=lstm_units, return_sequences=True
Flatten	Flatten	(None, pooled_length * 2 * lstm_units)	-	-
Dense	Fully Connected	(None, dense_units)	ReLU	units=dense_units
Dropout	Dropout2	(None, dense_units)	-	rate=dropout_rate
Output	Output Layer	(None, num_classes)	Softmax	units=num_classes

Table 2 Training configuration and hyperparameters used in the proposed model

Parameter	Value
Batch size	32
Optimizer	Adam
Dropout rate	0.3
Learning rate	0.001
Number of epochs	100
Loss function	Categorical Cross-Entropy
Activation function	ReLU (hidden layers), Softmax (output layer)
Early stopping	Enabled (patience = 10)

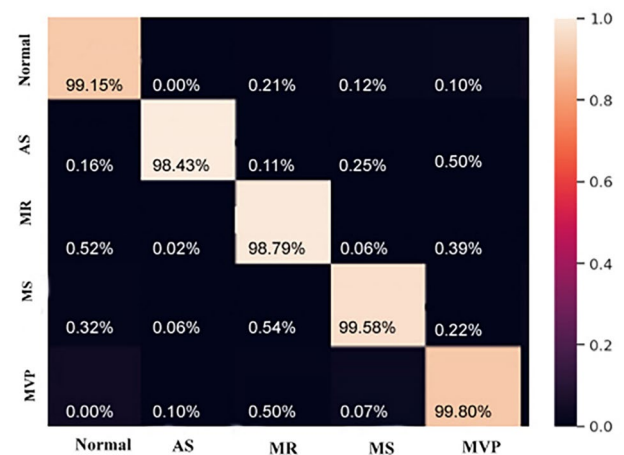
Table 3 Dataset split for training, validation, and testing across all five classes

Class Label	Training Samples	Validation Samples	Testing Samples	Total
NOR	320	80	80	480
AS	320	80	80	480
MR	320	80	80	480
MS	320	80	80	480
MVP	320	80	80	480
Total	1600	400	400	2400

Performance Results

Various experiments are conducted using window lengths from 0.031s to 2.011s to determine the ideal parameter. After evaluating SNR improvements, a window length of 0.8s was identified as providing a balanced trade-off between latency and effective denoising. The denoising model is trained using a dataset comprising 93,580 noisy and noise-free PCG frames, maintaining the original train-validation split of the dataset. A total of 2400 samples were used in this study. Each class is evenly represented in both subsets to ensure balanced learning and unbiased evaluation. This explicit partitioning supports reproducibility and ensures a fair assessment of the model's performance across all valvular heart disease categories.

The dataset used in this study was equally distributed across all five classes, with 480 samples per class. A stratified

**Fig. 6** Confusion matrix showing the classification performance of the proposed X-CBNet model across classes of heart sound signals

split was applied to divide the dataset into 1600 training samples (67%), 400 validation samples (17%), and 400 testing samples (17%). Due to the balanced nature of the dataset, no oversampling or data augmentation techniques were employed during model training. This approach ensures that each class was equally represented in all phases, reducing bias and enhancing the model's ability to generalize across categories and shown in Table 3.

In this research, the training phase involved a dataset of 1,600 images, evenly distributed across five categories: MR, MS, MVP, NOR, and AS, with 320 images per class. In the subsequent testing phase, the CNN-BiLSTM model demonstrated significant performance, achieving an accuracy (acc) of 99.15% on a separate test set containing 400 images. Figure 6 presents a detailed breakdown of the model's classification performance through the confusion matrix summarizing the performance of the proposed model on the five-class classification task. The model demonstrates high accuracy across all classes, with a normal 99.15%, AS 98.43%, MR 98.79%, MS 99.58% and MVP 99.80% correctly identified with minimal confusion. Notably,

the confusion between AS and MVP 0.50% and MR and MVP 0.39% indicates slight overlaps in feature patterns between these classes. Nevertheless, the model consistently maintains a misclassification rate below 1% across all categories, showcasing its robustness and discriminative capability. These results affirm that the proposed X-CBNet model can reliably distinguish between various VHDs and normal heart sounds, making it suitable for real-time diagnostic applications.

Table 4 presents the number and nature of misclassifications observed across all five classes, derived from the confusion matrix. Most misclassifications involve acoustically similar conditions such as AS and MR or MVP, suggesting overlap in learned spectral features.

The individual accuracies of each class are 99.15%, 98.43%, 98.79%, 99.58%, and 99.80% for normal, AS, MR, MS and MVP, respectively. The obtained accuracy and loss of the model are shown in Fig. 7 illustrates the training dynamics of the proposed X-CBNet model. Figure 7 (a) shows that both training and testing accuracy improve steadily over epochs, reaching above 99% by the end of training. The close alignment of both curves throughout the training process indicates that the model generalizes well without overfitting. Figure 7(b) depicts the loss curves, which show a sharp decline during the initial epochs and gradual convergence after 40 epochs. The final testing loss remains close to the training loss, further reinforcing the model's robustness and learning stability. The results confirm the model's capacity to learn effectively from the phonocardiogram signal data.

The AUC-ROC curve of the model performance is shown in Fig. 8. The ROC curves for each class and the aggregated micro and macro averages. The macro-average AUC reaches 0.99, while the micro-average AUC is 0.98, reflecting the model's consistent classification ability across all classes. Class-wise, the AUCs for NOR and MS are both 0.99, whereas AS, MR, and MVP maintain strong AUCs of 0.98. The close proximity of all curves to the top-left corner

Table 4 Misclassification summary based on the confusion matrix

True Class	Predicted as	Number of Misclassifications	Total Samples
AS	MVP / MR	2	80
MR	AS / MVP	1	80
MS	MR	0	80
Mitral Valve Prolapse (MVP)	AS / MR	1	80
NOR	AS / MVP	1	80
Total Misclassifications	—	5	400

of the graph highlights the high true positive rates and low false positive rates achieved by the model. These results affirm that the proposed architecture is not only highly accurate but also reliable across all categories of valvular heart disease.

Explainability of the Predictive Model

The study employs Grad-CAM, a specific XAI technique [21], to address the need for interpretability. This approach extracts gradients from the final convolutional layer and applies them to mel-spectrogram images derived from PCG signals. This process generates heat maps that emphasise significant regions. In the mel-spectrogram images of PCG signals as shown in Fig. 9, these critical regions predominantly highlight low-frequency patterns associated with PCG signals. The time-frequency analysis of a PCG signal through two key representations: the Mel spectrogram and MFCC. In the top panel, the Mel spectrogram reveals pronounced energy concentrations in the lower frequency bands (below 1000 Hz), which are characteristic of heart sounds such as S1 and S2. These concentrated regions are essential for identifying abnormal heart events. The bottom panel shows the MFCCs, which provide a compact and perceptually meaningful representation of the signal's spectral envelope. These features are particularly effective in modeling temporal variations and

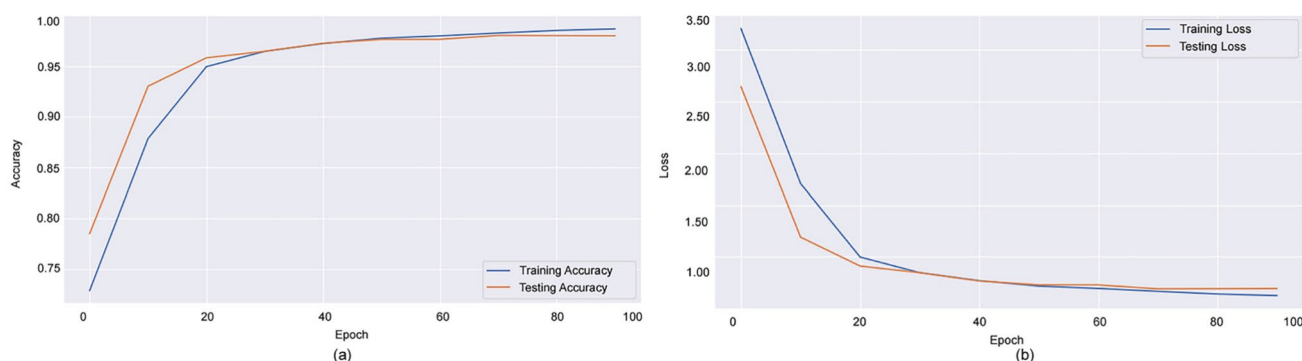


Fig. 7 (a) Training and testing accuracy. (b) Training and testing loss. The plots illustrate the learning behavior of the proposed X-CBNet model during the training phase

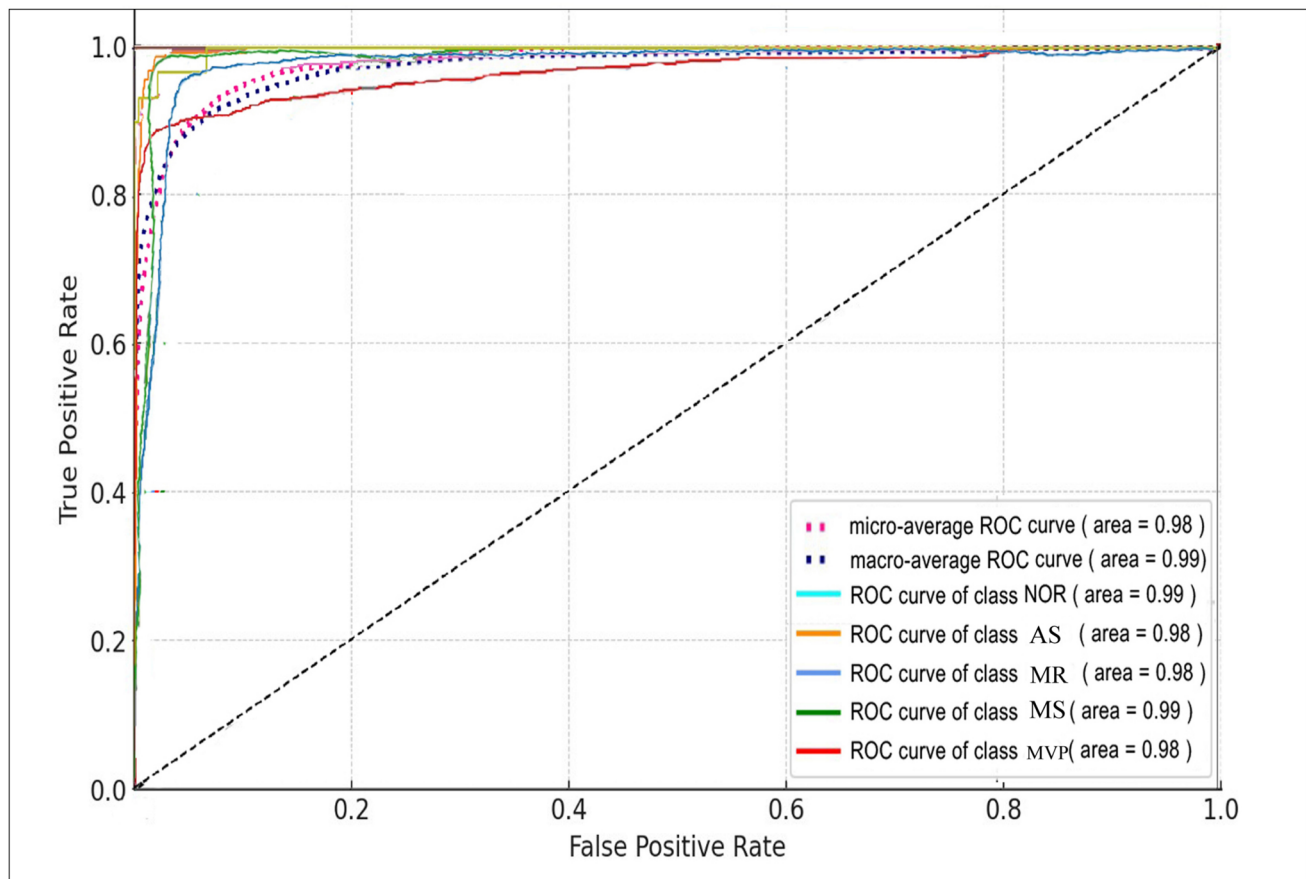
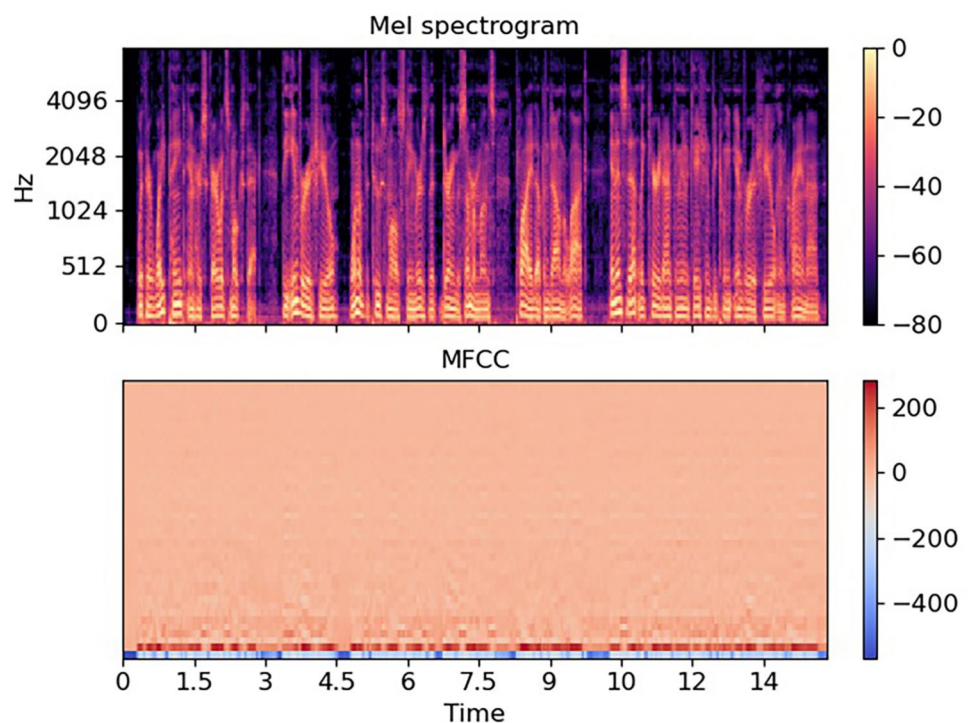


Fig. 8 ROC curves for all five classes. The AUC values for all classes exceed 0.98, indicating excellent classification performance by the proposed X-CBNet model

Fig. 9 Visualization of a heart sound signal using (top) Mel spectrogram and (bottom) MFCC. The MFCC spectrum across both temporal and spectral domains



are widely used in heart sound classification. Together, these visualizations validate the choice of input features used in the proposed X-CBNet model.

Identifying these low-frequency areas is crucial for precise classification, as they distinctly differentiate between various groups. The misclassifications are further analyzed using Grad-CAM, an XAI method that generates heatmaps to visually represent the regions in input images focused on during classification as shown in Fig. 10. The heat map shows the activation regions for a specific class in the input image. The bright areas (in yellow and red) indicate the regions of the image that the model considers important for classifying this input as the target class. The colour scale on the right of each heat map ranges from blue (low importance) to red (high importance). Regions with warmer colours (yellow and red) are more influential in the model's decision-making process. The activation regions are concentrated at specific points, indicating key features the model uses to classify the image into the normal class. The activation regions of the AS class are more diffuse compared to the normal class, with less intense highlights. This could indicate that the class represented by this heat map has more subtle distinguishing features, making it harder for the model to confidently classify. This may suggest a need

for more training data or feature engineering for better performance. The regions of the MR class are smaller but more numerous, scattered across the image. The model is picking up on multiple, less prominent features to make a decision. This pattern can be indicative of a complex class where multiple characteristics need to be considered for accurate classification. For the MS class, the activations are concentrated in larger areas, indicating broader regions of interest. Broad regions of activation can mean the model relies on a larger context within the image. This might be suitable for classes where overall patterns or textures are more important than specific features. For the MVP class, the heat map shows a combination of diffuse and concentrated regions.

The heatmaps prominently highlight the low-frequency regions of the mel-spectrogram image. For instance, the Grad-CAM image depicting AS misclassified as MR resembles the heatmap seen for the true MR class, suggesting the model focused on features reminiscent of MR characteristics, leading to misclassification. Conversely, the Grad-CAM visualization for MR is misclassified as AS. It clearly emphasizes features typical of the AS class rather than MR. Similar patterns are observed for MVP and N, with instances of misclassification showing distinct heatmap characteristics corresponding to different classes. This

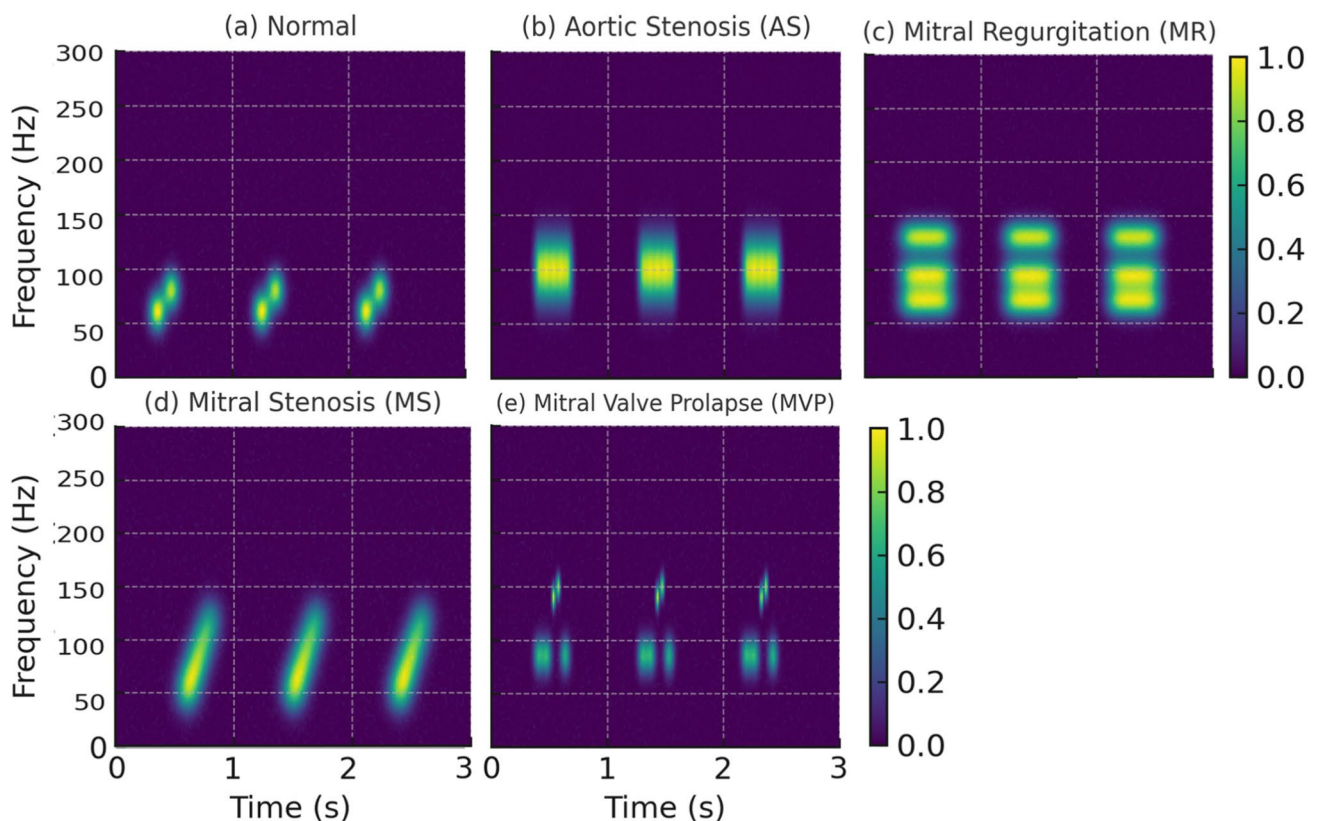


Fig. 10 Grad-CAM heatmaps for five heart sound classes. The visualizations highlight the regions of the mel-spectrograms that the X-CBNet model considers most influential during classification. The

red and yellow areas denote higher attention by the model, indicating class-specific discriminative features

Table 5 Comparison of proposed X-CBNet with existing heart sound classification methods

Author (Year)	Method	Dataset	Acc	Sen	Spec	AUC	Explainability
Sundar et al. [27]	LSTM	PhysioNet /CinC	94.5%	—	—	—	No
Riccio et al. [24]	Binary CNN	Private PCG	85%	—	—	—	No
Ismail et al. [14]	Hybrid CNN-SVM	PhysioNet PCG	95%	93%	96%	—	No
Carter et al. [8]	CNN + SHAP	PhysioNet PCG	78%	—	—	0.85	Yes (SHAP)
Al-Issa and Alqudah [2]	CNN-LSTM	PhysioNet PCG	98.5% (non-aug) / 99.87% (aug)	—	—	0.9978 / 0.9985	No
Alkhodari and Fraiwan [4]	CNN-BiLSTM	PhysioNet PCG	78.31%	—	—	—	No
Shuvo et al. [26]	CRNN	PhysioNet PCG	86.57%	—	—	—	No
Proposed	X-CBNet	PhysioNet /CinC	99.15%	99%	98%	0.99	Yes (Grad-CAM)

visualization technique elucidates how the model operates by showing which image areas receive greater attention during prediction, aiding in the verification of the model's focus on relevant features for accurate predictions.

Comparison with State-of-the-art Methods

The performance of the proposed approach outperforms the conventional methods as shown in Table 5. The proposed model consistently achieves higher accuracy compared to other methods.

Discussion

The high classification accuracy and explainability of the proposed X-CBNet model make it well-suited for integration into real-time clinical workflows and telemedicine platforms. One potential use case involves embedding the model into electronic stethoscopes or mobile diagnostic tools used by primary healthcare workers. During patient auscultation, heart sound signals can be recorded and converted into mel-spectrograms on-device or via cloud services, with the X-CBNet model providing instant classification and visual explanation via Grad-CAM heatmaps. These results can assist general practitioners in making timely referrals to cardiologists, especially in rural or underserved areas.

Additionally, in telemedicine environments, PCG signals acquired at remote centers can be uploaded and analyzed via a secure web-based interface, where clinicians can access not only predicted classes but also visual justifications for each decision. This enhances clinical trust and supports decision-making without the need for direct specialist involvement. Such integration supports early screening, reduces unnecessary hospital visits, and facilitates remote cardiac monitoring. To support clinical deployment, future work may include packaging the model into a lightweight inference API compatible with embedded systems, and validating it through clinical trials involving diverse patient populations and real-world noise conditions.

In comparison with previous works, which typically reported accuracies in the range of 78–98% without offering interpretability, our proposed X-CBNet achieved a superior accuracy of 99.15% and a macro-average AUC of 0.99. Unlike most black-box models, X-CBNet integrates Grad-CAM, providing visual explanations that enhance clinical trust and reliability. These results establish X-CBNet as a state-of-the-art framework for valvular heart disease classification using PCG signals.

Limitations

The dataset, limited to single labels, covers only a fraction of detectable conditions through sound, indicating a need for expanded research across a wider range of conditions. The size of the multi-label dataset is relatively modest compared to the extensive patient audio data, encompassing multiple diseases per case. Increasing the dataset size would enhance the selection of neural network architectures and SHAP parameters in future studies. Moreover, our experimental findings require additional validation from healthcare professionals. Depending on the application, physicians or researchers may opt for alternative conditions or utilize a larger dataset to improve model accuracy and tailor its application to specific clinical or research objectives. Finally, the small number of misclassifications observed between acoustically similar classes like AS and MR suggests a need for more discriminative feature representations. Future research could investigate integrating domain-specific handcrafted features or multi-view learning strategies to mitigate such overlaps.

Another limitation is that the PhysioNet/CinC 2016 dataset, while covering patients of varying ages and genders, does not provide detailed demographic distributions, which restricts deeper analysis of how factors such as age or sex may influence PCG characteristics and classification outcomes. Another important aspect to consider in future work is the occurrence of multi-label cases, where a single patient may present with more than one valvular disorder.

simultaneously. The current study focused on single-label classification, but extending the framework to handle multi-label scenarios would more closely reflect clinical reality and enhance the applicability of automated PCG analysis.

Conclusion

This study underscores the importance of accurate and timely diagnosis in addressing VHDs, a significant global health challenge. Using PCG signals, the proposed X-CBNet model trained on mel-spectrogram images derived from these signals has demonstrated 99.15% accuracy in classifying VHDs. To interpret the proposed model, an XAI technique Grad-CAM is employed that visualizes the model's decision-making process by highlighting its focus areas. This transparency could enhance the model's utility in real-time clinical settings. The integrated approach is promising in advancing VHD detection, facilitating more informed medical interventions, and improving patient outcomes. Future research will leverage larger datasets and explore various XAI techniques across all model layers to enhance interpretability.

Abbreviations

<i>VHD</i>	Valvular Heart Disease
<i>CNN</i>	Convolutional Neural Network
<i>Bi-LSTM</i>	Bidi-irectional Long Short-Term Memory
<i>PCG</i>	Phonocardiograms
<i>Grad-CAM</i>	Gradient-weighted Class Activation Mapping
<i>AI</i>	Artificial Intelligence
<i>AR</i>	Aortic Regurgitation
<i>AS</i>	Aortic stenosis
<i>MR</i>	Mitral Regurgitation
<i>MS</i>	Mitral Stenosis
<i>MVP</i>	Mitral Valve Prolapse
<i>DL</i>	Deep Learning
<i>XAI</i>	Explainable Artificial Intelligence
<i>RNN</i>	Recurrent Neural Network
<i>DFT</i>	Discrete Fourier Transform
<i>DWT</i>	Discrete Wavelet Transform
<i>MFCC</i>	Mel Frequency Cepstral Coefficient
<i>RMS</i>	Root Mean Square
<i>SNR</i>	Signal-to-Noise Ratio
<i>DNN</i>	Deep Neural Network
<i>GAP</i>	Global Average Pooling

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

Ethics Approval and Consent to Participate Ethics approval and consent to participate were not required for this study as it exclusively involved publicly available data, which adheres to all ethical standards.

Consent for Publication All authors have reviewed and approved the final version of the manuscript and consent to its publication.

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